

CBSE Class 10 Mathematics

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains questions. All questions are compulsory.
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section D: Long Answer (4-5 marks each)

1. Two water taps together can fill a tank in $9\frac{3}{8}$ hours. The tap of larger diameter takes 10 hours less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank. (hard)
2. If the roots of the quadratic equation $(a - b)x^2 + (b - c)x + (c - a) = 0$ are equal, prove that $2a = b + c$. (hard)